qTPU: Hybrid Tensor Networks for Quantum-Classical Acceleration

Paper # 1414

Abstract

Classical accelerators (GPUs, TPUs) excel at high-performance computing but cannot efficiently represent problems that scale exponentially, such as highly entangled quantum states. Quantum processors (QPUs) address these intractable problems but are constrained by noise, high error rates, and low throughput. This necessitates hybrid quantum-classical execution: computationally complex kernels are offloaded to QPUs, while scalable computation runs on classical accelerators.

Despite this clear need for hybrid systems, we currently lack a concise programming paradigm for expressing and optimizing them. Developers must manually partition and orchestrate applications across quantum and classical hardware through *ad hoc* approaches that produce fragmented, inflexible workflows. This prevents holistic optimization across the quantum-classical interface and limits scalability.

To this end, we introduce the *hybrid tensor network (hTN)* abstraction—a unified representation capturing quantum-classical computation—realized in qTPU, an end-to-end system comprising: the *qTPU programming model* for declarative hybrid computation specification; the *qTPU compiler* for holistic hTN optimization balancing classical cost against quantum error; and the *qTPU runtime* for scalable execution across heterogeneous QPUs and classical accelerators.

Our evaluation demonstrates that qTPU achieves 3–4 orders of magnitude lower classical overhead, up to 7.2× lower quantum error rates, up to 53× faster compilation, and over 20× end-to-end speedups across hybrid ML, circuit knitting, and quantum error mitigation.

1 Introduction

Data centers are increasingly relying on specialized accelerators, such as GPUs, TPUs, and FPGAs, to meet the demands of growing domain-specific compute workloads, *e.g.,* in machine learning and scientific computing [3, 42, 43, 65]. Through recent remarkable engineering efforts, quantum processors (QPUs) are becoming available as new domain-specific accelerators [2, 5, 29, 37]. When combined with classical computing, QPUs could help us address problems in materials science and chemistry that remain intractable on classical computers [38, 51, 63, 69, 74].

QPUs leverage quantum-mechanical properties such as superposition and entanglement to efficiently represent and manipulate exponentially large state spaces that cannot be encoded using classical memory [55]. This capability enables, *e.g.,* the simulation of strongly correlated quantum systems and the modeling of highly entangled or high-dimensional distributions in machine learning, and offers the potential for exponential speedups for certain problems [35, 72, 77].

Although QPUs offer unique advantages for specific classes of problems, they are subject to severe limitations: they are noisy, provide only limited computational resources in memory and availability of devices, and have throughput orders of magnitude lower than classical accelerators [39, 64].

As a result, quantum advantages – where they exist – typically apply only to the complex core of a computational problem, where classical devices fail due to the exponential complexity of representing quantum states [12, 39]. Many important workloads, such as complex machine-learning models or scientific simulations for material sciences, fall into a regime where they are both *(i)* too inefficient or outright infeasible to execute solely on classical accelerators, and *(ii)* too complex and large-scale to run entirely on today's QPU hardware [52, 63].

This motivates a *hybrid* quantum-classical execution paradigm. Instead of running a program entirely on classical hardware or entirely on a QPU, realistic applications distribute work across both: the quantum-inclined components are offloaded to QPUs, while the remaining computation runs on better-suited classical accelerators [12, 33].

To program such hybrid applications, the current approach relies on a rigid host–kernel programming model. Programmers must manually isolate portions of the algorithm that could benefit from quantum or classical acceleration and implement them as separate kernels [8, 57].

Unfortunately, this existing form of quantum–classical hybrid computing is tedious, inflexible, and inefficient to use in cloud environments, suffering from three core limitations:

(1) **Fragmented programming models:** Developing hybrid applications requires reasoning across two distinct programming paradigms and manually implementing separate routines for classical devices and QPUs [19]. This fragmented model forces developers to explicitly manage low-level data movement, control flow, and upfront partitioning decisions, making the development of hybrid applications tedious and prone to inefficiencies.

(2) **Lack of holistic compiler optimizations:** Current programming models require developers to define a fixed boundary between quantum and classical kernels *a priori* [8]. This rigid separation prevents compilers from reasoning across the quantum–classical interface, hindering holistic, end-to-end optimization. As a result, no dynamic execution decisions are possible; *e.g.,* even if parts of a quantum kernel could be executed more effi-

ciently on a classical accelerator, they cannot be automatically offloaded, locking the application into a potentially suboptimal execution path [83].

(3) **Poor scalability and rigid runtime:** Current hybrid workflows are inefficient and struggle to scale because they are manually orchestrated and executed as separate, sequential steps, lacking a unified program view [32]. This prevents dynamic execution decisions, *e.g.,* even if part of a quantum kernel would run more efficiently on a classical accelerator, the compiler cannot offload it automatically.

We argue that these limitations fundamentally arise from the lack of a concise representation that seamlessly captures both quantum and classical computation, leading to our main research question:

Research question

How can we design a concise and expressive hybrid computing paradigm that unifies quantum and classical computation while enabling efficient execution across heterogeneous accelerators?

Our key idea for addressing this challenge is the **hybrid Tensor Network (hTN)** abstraction, a necessary theoretical foundation to address the fragmentation in hybrid computing. In particular, we establish Tensor Networks (TNs) [58]—a core algebraic formalism for both quantum circuit simulation and accelerated classical computation—as a unifying computational substrate [50, 86]. The hTN extends this formalism by representing the complete hybrid workflow, including quantum components (quantum tensors) and classical components (classical tensors), as a single, connected graph. This unified expression bridges the classical-quantum gap, enabling holistic optimization and treatment of complex algorithms, such as hybrid machine learning models, as a single, coherent computational entity.

To realize hTNs in practice and to answer our posed research question, we present qTPU, an end-to-end systems approach for unified, flexible and scalable hybrid computing. *(1)* Our *qTPU Programming Model* formally defines the primitives built on hTNs, allowing programmers to specify only *what* to compute, *not how*; our *(2) qTPU Compiler* converts hTNs into executable hybrid code, which we holistically optimize for efficient computation across a set of QPUs and classical accelerators; and finally, *(3)* our *qTPU Runtime* dynamically adapts the hybrid computation to scalably execute across the available heterogeneous collection of classical and quantum accelerators.

Our evaluation shows that qTPU achieves substantial improvements across hybrid workloads: up to 53× faster compilation, 7.2× lower quantum error rate, 3,700× smaller generated code, and 10,000× lower classical overhead than state-of-the-art baselines. qTPU further scales to problem sizes that existing hybrid systems cannot handle, while enabling efficient utilization of heterogeneous QPU/GPU resources.

Contributions. We make the following contributions.

- We introduce the hybrid Tensor Network (hTN) abstraction, a unified representation that captures hybrid quantum-classical workflows as a single connected tensor graph, moving beyond existing host-kernel models (§ 3.2).
- We formally introduce the qTPU Programming Model, which realizes our hTN abstraction. We define its core primitives—including the iswitch (for specifying quantum tensors) and hEinsum (for hybrid contraction)—which enable the simple and declarative description of entire hybrid quantum-classical programs (§ 5).
- We introduce compiler transformations that jointly optimize across both modalities by balancing the classical cost of tensor contractions with the quantum execution error, ensuring optimal utilization of available quantum and classical compute paradigms. (§ 6).
- We design a scalable execution approach that distributes and dynamically adapts hybrid programs across a heterogeneous set of QPUs and classical accelerators. The qTPU runtime facilitates this through runtime code generation, tailoring device-specific kernels on demand to maximize parallel execution and efficient resource utilization based on available hardware. (§ 7).

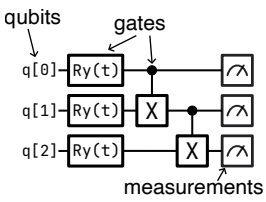
2 Background and Motivation

2.1 Foundations of Quantum Computing

Quantum computing could accelerate tasks in many areas, including quantum physics, materials science, and machine learning [35, 38, 69, 77]. Although current QPUs remain in their early stages, recent experiments have demonstrated behavior that is increasingly difficult to reproduce classically, suggesting potential advantages for specific problem classes [4, 16, 17, 36, 79].

QPUs are typically programmed under the *quantum kernel model*, illustrated in Fig. 1. A quantum kernel is a classical function that produces a *quantum circuit*, which is then compiled and executed on a QPU [57].

The basic computational unit of a quantum circuit is the *qubit*, a two-level quantum system. An n-qubit register corresponds to a complex-valued vector in a 2^n-dimensional Hilbert space, enabling it to represent vectors that would require exponentially many parameters to store classically [55]. A quantum circuit generally *(1)* initializes qubits, *(2)* applies single- and multi-qubit quantum gates, and *(3)* measures a subset of the qubits in the computational basis. Measurement collapses each qubit to either 0 or 1 according to a distribution determined by the circuit; repeating the circuit for s *shots* yields bitstrings $b^{(\ell)} \in \{0,1\}^m, \ell = 1, \ldots, s$, over the m measured qubits.




```
@qkernel
def simple_kernel(t)
    q = alloc_qubits(3)
    for i in range(3):
        ry(q[0], t)
    cx(q[0], q[1])
    cx(q[1], q[2])
    measure(q)

circuit = simple_kernel(0.3)
res = qpu.run(circ, shots=1000)
```

Figure 1: Example of the quantum-kernel model.

2.2 Quantum-Classical Computing

Both quantum and classical accelerators can efficiently solve domain-specific problems; however, neither provides a one-size-fits-all solution.

On the one hand, modern *classical accelerators* (such as GPUs, TPUs) provide massive data and operation parallelism, large on-device memory, and extremely high throughput and low latency for, *e.g.,* dense numerical workloads. They are reliable, abundantly deployed, and scale to thousands of devices in distributed cloud environments. However, classical accelerators operate on data structures that cannot efficiently represent strongly correlated and entangled quantum systems, whose classical representations scale exponentially compared to their quantum representations [28, 30, 93].

On the other hand, *quantum accelerators* (QPUs) leverage unique quantum-mechanical effects to efficiently represent and manipulate highly correlated quantum states that classical systems cannot handle [39, 55]. This makes them attractive for tasks such as simulating highly correlated systems for material science or quantum chemistry. However, QPUs also have inherent limitations, since qubits are inherently noisy, and quantum operations are prone to errors, resulting in unreliable results at large circuit depths [61, 71]. Achieving fault tolerance requires substantial overhead in both qubit resources and execution time, making logical qubits a scarce resource and significantly limiting QPU throughput and latency.

The need for hybrid acceleration. Classical and quantum accelerators have complementary strengths and limitations, making neither suitable for all tasks. Realistic workloads therefore require a tightly integrated hybrid approach: QPUs are used selectively where quantum capabilities offer an advantage, while all other computation is handled by classical accelerators.

2.3 Blueprint of Hybrid Applications

Hybrid quantum-classical computing leverages the complementary strengths of classical accelerators and QPUs to solve problems that neither can efficiently address on its own.

The *core* of virtually all hybrid quantum-classical algorithms and methods—including quantum simulation [54, 59, 91], circuit knitting [80, 85], error mitigation [48, 84], and quantum-enhanced machine learning (ML) [10, 14, 49]—

```
1  @qkernel # QNN kernel (input x, weights W, output o)
2  def quantum_layer(x, W, o):
3      q = alloc_qubits(N)
4      for i in range(N):
5          ry(q[i], x[i]) # input encoding
6      model(q, W) # apply some QNN-layer
7      apply_obs(q, o) # measure specific observable
8      measure(q)
9
10 @ckernel # a classical GPU/TPU kernel
11 def classical_layer(Z, V): # linear layer
12     return matmul(V, Z) # matrix-matrix multiplication
13
14 def forward(X, W, V, O): # hybrid model evaluation
15     # Stage 1: construct and run a family of circuits
16     circuits = [quantum_layer(x,W,o) for x in X for o in O]
17     Z = qpu.run(circuits, shots=1000)
18     # Stage 2: perform classical processing
19     Y_pred = classical_layer(Z, V)
20     return Y_pred
21
22 def train(): # model training loop
23     X,Y = dataset(); W,V = init_weights(); O = observables()
24     for epoch in range(E):
25         Y_pred = forward(X, W, V, O)
26         update([W, V], loss(Y, Y_pred))
```

Listing 1: Fragmented quantum-classical ML example.

shares a common computational blueprint, consisting of two tightly coupled stages. For example, Lst. 1 shows a quantum-classical approach to train a machine-learning kernel, implemented via the kernel's `forward`-pass.

Stage 1: Evaluate families of quantum circuits. The workflow begins by executing a *family* of closely related quantum circuits derived from classical inputs or intermediate computation. These circuits typically share a structural backbone but differ in controlled ways, together exposing the necessary information required for the hybrid computation.

In our example (Lst. 1, Line 16-17), we produce and run a family of quantum circuits that encode the individual inputs of our training dataset and measure observables, whose results are later fed into the next layer. This family needs to be explicitly generated and run on a specific QPU to produce all the results required for the next stage.

Stage 2: Perform classical processing. The measurement outcomes from the circuit family are then combined with additional classical inputs and processed using classical computation. Across algorithmic classes, this postprocessing is governed almost entirely by *linear-algebraic* operations, which transform the quantum-generated data into the quantities of interest and feed back into the host-driven control loop that produces the next family of circuit evaluations.

In our example (Lst. 1, Line 19), we apply a simple linear layer—*e.g.,* on a GPU—to the quantum outputs and classical weights, producing the predicted results for the dataset.

Examples of hybrid applications. Beyond our example of hybrid machine learning, various other applications in quantum computing follow the same structure. In this work, we focus on three representative hybrid applications:

(i) **Hybrid Machine Learning** combines variational quantum circuits with classical layers to model functions or data distributions that cannot be efficiently captured by purely classical or purely quantum models [10], similar to our toy-example of Lst. 1;

(ii) **Scalable hybrid computing** uses circuit knitting techniques [80,85] to execute circuits that exceed the qubit capacity of QPUs by decomposing a large circuit into families of smaller subcircuits, running them on small QPUs, and classically reconstructing the result; and

(iii) **Quantum Error Mitigation (QEM)** generates and executes families of circuit variations to estimate and suppress noise effects, combining the resulting measurements through classical postprocessing to approximate the noiseless output of the original circuit [26,47,84].

We evaluate our approach on these applications in § 8.

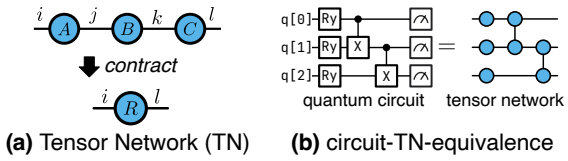
2.4 Limitations of the Current Practices

Using both quantum and classical resources leverages the full benefits of both worlds. Unfortunately, the current approach to writing quantum-classical programs follows a manual, ad-hoc workflow that is error-prone and complex due to several fundamental limitations, which we explain now.

Limitation #1: Fragmented programming model. In current hybrid workflows [8, 57], the quantum and classical parts of a program must be written in separate paradigms, as shown in Lst. 1. The programmer implements one kernel for the QPU and another for the GPU/TPU, and must manually coordinate them in the host code. In the example, the `forward` function builds a family of circuits, sends them to the QPU, waits for results, and then invokes a classical kernel. Thus, even though the programmer specifies *what* the computation should do, they must also express *how* the quantum and classical kernels interact, making hybrid programs unnecessarily complex.

Limitation #2: Inflexible compilation. The programmer must fix the quantum–classical boundary a priori, preventing holistic cross-paradigm optimization because the hybrid computation is not represented as a single unified object. In Lst. 1, the quantum and classical kernels are statically tied to their modalities—QPUs for `quantum_layer` and GPUs/TPUs for `classical_layer`. This rigidity blocks optimization of `forward` as a whole, since it consists of two separately compiled black-box kernels. As a result, tasks cannot be reassigned across paradigms, even when parts of `quantum_layer` could run more efficiently on a classical accelerator.

Limitation #3: Rigid runtime. Current hybrid workflows are manually orchestrated and executed as separate, sequential steps, giving the runtime no unified view of the computation. In Lst. 1, the manual orchestration inside `forward` fixes the quantum and classical stages in order and binds them to their respective devices, preventing dynamic parallelism or resource-aware scheduling. Because the assignment of work is static, the runtime cannot adapt to the availability or load of heterogeneous QPUs and classical accelerators, which severely limits scalability for larger problems.



(a) Tensor Network (TN) **(b)** circuit-TN-equivalence

Figure 2: Tensor Network background (§ 3.1).

Problem statement

We require a solution that unifies quantum-classical applications in a common foundation, thus allowing holistic optimization across both fragments, while adapting its execution to available quantum-classical devices.

3 Hybrid Tensor Networks (hTNs)

We propose *hybrid tensor networks* (hTN), which address the above limitations by unifying quantum and classical computing in a common framework.

3.1 Tensor Networks (TNs)

Tensor networks compactly describe an output tensor as a function of input tensors.

Einsum expression. TNs specify pure linear-algebra expressions via the *Einsum* language. Formally, $T_O = \mathsf{einsum}(I_1, \ldots, I_n \to O;\ T_1, \ldots, T_n)$ defines the output tensor T_O entry-wise as

$$T_O[\mathbf{o}] \triangleq \sum_{\mathbf{i} \in (\mathcal{I} \setminus O)} \prod_{k=1}^{n} T_k[\mathbf{i}_{I_k}], \tag{1}$$

where T_i are tensors with index set I_i; O is the output index set, of which $\mathbf{o}$ denotes an assignment; $\mathcal{I} = \bigcup_{k=1}^{n} I_k$ is the union of all index sets; and $\mathbf{i}_{I_k}$ is the restriction of the global index assignment $\mathbf{i}$ to the indices of tensor T_k. For example,

$$R = \mathsf{einsum}(ij, jk, kl \to il;\ A, B, C) \tag{2}$$

specifies the triple matrix product $R = A \cdot B \cdot C$, *i.e.,* $R_{il} = \sum_{j,k} A_{ij} B_{jk} C_{kl}$.

Tensor networks as graphs. An equivalent way to specify a TN is as an undirected graph, whose vertices represent tensors and the edges correspond to shared indices (*i.e.,* contractions). When two tensors are connected by an edge, the corresponding index is contracted (summed over), and dangling edges connected to only a single vertex specify the output indices. Fig. 2 (a) shows an example of such a TN-graph, equivalent to the einsum expression of Eq. 2.

Quantum computations as TNs. The connection between quantum and classical computation becomes explicit when we observe that any quantum circuit can be represented as a TN, as we show by example of Fig. 2 (b). Thus, quantum circuits can be expressed using the einsum-based formalism introduced above, enabling a unified representation of both classical and quantum computations.

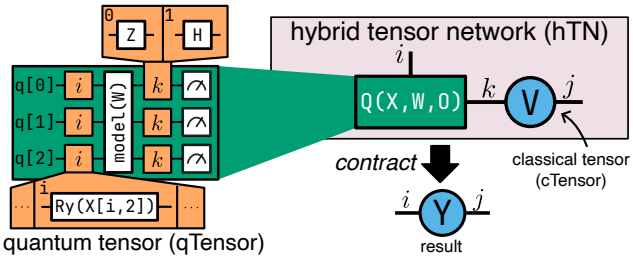


Figure 3: Our hybrid tensor network (hTN) abstraction (§ 3.2).

3.2 Key Idea: Hybrid Tensor Networks (hTNs)

We introduce the **hybrid tensor network (hTN)**, an abstraction that extends the TN formalism to enable unified execution and flexible partitioning of computation across quantum and classical accelerators, yielding a simple and coherent abstraction for hybrid quantum-classical workloads.

Fig. 3 shows an hTN describing the computation of the hybrid `forward` function in Lst. 1. The program executes several circuits over a batch of input data $X \in \mathbb{R}^{|i|\times 3}$ containing $|i| = d$ data points, each represented as a vector of dimension 3, and a set of observables O with $|O| = 2$. The qTensor on the left in Fig. 3 compactly describes this circuit's entire family. The qTensor has two indices, i and k, with sizes $|i| = d, |k| = 2$. These indices are referenced by *index-instructions*, associating each with several operations. For instance, instantiating i=1,k=0 replaces every index-instruction that depends on i with the second operation in its family, and every index-instruction of k with the first operation in its family.

The hTN on the right in Fig. 3 compactly describes the hybrid program with the qTensor and a cTensor $V \in \mathbb{R}^{|k|\times|j|}$. The index k contracts with the qTensor's output index, while j denotes the dimension of the output vectors $y \in Y$. Contracting this hTN yields the result tensor $Y \in \mathbb{R}^{|i|\times|j|}$, corresponding to the output of the hybrid computation and containing the resulting ML-layer predictions for each input indexed by i.

Semantics. Unlike classical TNs, an hTN first executes its qTensors to its results-cTensor, before contracting the hTN to a tensor with einsum. An hTN, consisting of classical tensors $T_1^c, \ldots, T_n^c$, each with its own index set I_i^c, and qTensors $T_1^q, \ldots, T_m^q$, each with its own index set I_i^q, we contract as follows:

$$T_O = \text{einsum}(I_1^c, \ldots, I_n^c, I_1^q, \ldots, I_m^q \rightarrow O; T_1^c, \ldots, T_n^c, [\![T_1^q]\!], \ldots, [\![T_m^q]\!]),$$

where $[\![T^q]\!]$ denotes executing the qTensor for all its indices and transferring the results to a cTensor.

4 QTPU System Overview

To realize our hTN abstraction, we introduce QTPU, an end-to-end system that enables programmable and scalable hybrid computations. Fig. 4 shows an overview of the system.

To express hybrid programs with the hTN abstraction, users program in the **QTPU programming model**, which provides a simple, declarative interface based on the two primitives iswitch and hEinsum, while entirely hiding the complexity of coordinating quantum and classical computation.

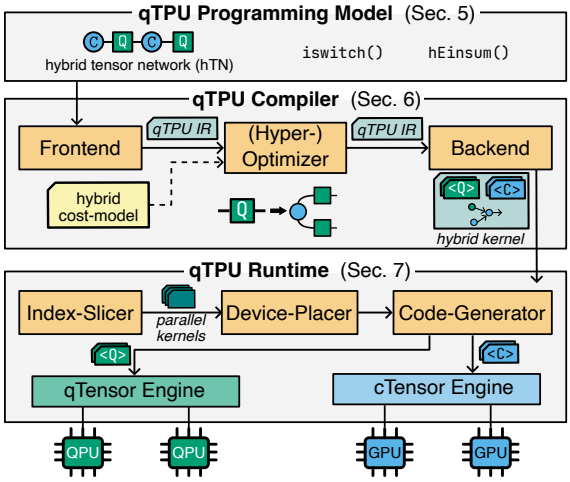


Figure 4: System overview (§ 4).

To automatically co-optimize a hybrid program, the **QTPU compiler** optimizes it to fully utilize both quantum and classical computing. The compiler's frontend first parses the program into the unified QTPU IR, after which the optimizer applies a hybrid quantum-classical cost model to identify Pareto-optimal rewrites that reduce quantum errors from noise and classical costs in terms of operations and memory usage. Finally, the backend produces an optimized hybrid kernel comprising device-agnostic quantum and classical kernels, along with a canonical operation schedule, ready for execution.

Finally, the **QTPU runtime** scalably executes the optimized hybrid program, adapting it to the available resources. To do so, its index slicer first partitions the hTN into smaller, independent, and parallelizable slices. The device placer then maps each concrete quantum and classical computation to an available device to maximize throughput, after which the code generator produces device-optimized kernel code for each computation. The hTN is then executed in a distributed pipeline, where the qTensor engine evaluates all qTensors on the available QPUs, and the cTensor engine contracts the resulting cTensors using the available classical accelerators.

5 QTPU Programming Model

We propose the QTPU programming model, which provides a unified abstraction for hybrid quantum-classical computation, enabling programmers to state hybrid computations *declaratively*, without their low-level execution details – such as manually stitching components into a single workflow.

Our QTPU model exposes two core constructs: (1) *quantum tensors*, created by annotating quantum programs with iswitch statements, and (2) a declarative multi-tensor contraction operator, hEinsum, which composes classical and quantum tensors into a single hybrid computation. These primitives are in addition to the classical tensor (cTensor)

primitive, represented by a multi-dimensional array.

5.1 qTensor Primitives

A quantum tensor (qTensor) compactly represents a multi-index family of quantum kernels. It is structured as a quantum program annotated with iswitch operations, each binding a symbolic index to a discrete set of circuits. We consider a simple language t^Q of defining quantum circuits, consisting of gates and our novel iswitch operator:

$$t^Q := \text{gate}(G, \overline{q}) \mid \text{seq}(t^Q, t^Q) \mid \text{iswitch}(k, \overline{G}, \overline{q})$$

Here, G references a quantum gate, while $\overline{G}$ is a vector of gates, and seq refers to a sequence of instructions. $\overline{q}$ is a vector of unique qubit identifiers.

Materialization. Instead of executing the qTensor on a QPU, we could *materialize* it to a classical tensor, given an index set $\mathcal{I}$. U_G is the unitary transformation of gate G.

$$\begin{aligned}
[\![\text{gate}(G, \overline{q})]\!]^{\text{m}}_{\mathcal{I}} &= U_G \\
[\![\text{seq}(t^Q_1, t^Q_2)]\!]^{\text{m}}_{\mathcal{I}} &= [\![t^Q_1]\!]^{\text{m}}_{\mathcal{I}} \cdot [\![t^Q_2]\!]^{\text{m}}_{\mathcal{I}} \\
[\![\text{iswitch}(k, \overline{G}, \overline{q})]\!]^{\text{m}}_{\mathcal{I}} &= U_{\overline{G}_{\mathcal{I}_k}}
\end{aligned}$$

5.2 hEinsum: Hybrid Einsum Primitive

We consider another language t^T for tensor computations, embedding quantum computations (t^Q) and our novel hEinsum operator. hEinsum *declaratively* specifies a hybrid TN operation, thus relieving the programmer from managing execution details, while enabling the compiler and runtime to execute it efficiently on QPUs and classical accelerators.

Operationally, hEinsum acts like a standard TN contraction over classical tensors, where qTensors are materialized first to cTensors. To formally state the semantics of our hEinsum term, we define our simple tensor language:

$$t^T := \text{cTensor}(T) \mid \text{qTensor}(t^Q) \mid \text{hEinsum}(\text{expr}; t^T_1, \ldots, t^T_n)$$

We can define the semantics of that language, in particular our hEinsum term, using the mathematical einsum operator (Sec. 3.2). Intuitively, we materialize any qTensor to its classical counterpart, over which we compute a classical einsum.

$$\begin{aligned}
&[\![\text{cTensor}(T)]\!]^{\text{s}}_{\mathcal{I}} &&= T \\
&[\![\text{qTensor}(t^Q)]\!]^{\text{s}}_{\mathcal{I}} &&= [\![t^Q]\!]^{\text{m}}_{\mathcal{I}} \\
&[\![\text{hEinsum}(I_1, \ldots, I_n \rightarrow O; T_1, \ldots, T_n)]\!]^{\text{s}}_{\mathcal{I}} &&= \\
&\qquad \text{einsum}(\text{expr};\ [\![T_1]\!]^{\text{s}}_{I_1}, \ldots, [\![T_n]\!]^{\text{s}}_{I_n}) \text{ where } \textstyle\bigcup_{k=1}^{n} I_k \subseteq \mathcal{I}
\end{aligned}$$

Summary. The QTPU programming model provides a simple, declarative framework for hybrid quantum-classical computation. QTensors introduce indexed quantum kernels via iswitch annotations, while hEinsum lifts classical einsum semantics to the hybrid setting. By deferring kernel specialization and execution until contraction time, the model separates logical specification from concrete realization, permitting the compiler and runtime to optimize placement and scheduling across QPUs and classical accelerators.

5.3 Example Continued

To demonstrate the versatility of the QTPU programming model, we revisit the hybrid ML task from Lst. 1, reimplemented in Lst. 2 (visualized in Fig. 3). The `quantum_layer` function (Line 2) defines a two-dimensional qTensor encoding the required circuit family: index `i` iterates over dataset X to select the input encoding, while `k` selects the measurement observable.

```
1  @quantum_tensor
2  def quantum_layer(X, W, O):
3      q = alloc_qubits(N)
4      for i in range(N):
5          # batch-dimension index
6          iswitch("i", [ry(x[i]) for x in X], q[i])
7      model(q, W)
8      # hidden-dimension index
9      iswitch("k", [apply_obs(o) for o in O], q)
10     measure(q)
11
12 def forward(X, W, V, O): # hybrid computation (declarative)
13     return hEinsum("jk,ik->ij", V, quantum_layer(X, W, O))
```

Listing 2: Unified Quantum-Classical ML example (cf. Lst. 1)

The iswitch primitive instantiates operations based on `i` and `k`. For instance, `i=1,k=0` maps to the second data point and first observable. This allows the programmer to declaratively specify the complete circuit family within the qTensor function, avoiding manual construction of individual circuits.

The hybrid computation in the `forward` function is realized via hEinsum, which defines the contraction over quantum and classical tensors (cf. Fig. 3). In this model, the programmer specifies the computation solely through tensor structure, while details of circuit scheduling, data movement, and coordination remain entirely hidden.

6 QTPU Compiler

The QTPU compiler lowers the high-level, declarative hEinsum-description (§ 5) into an optimized hybrid kernel.

Workflow. Fig. 5 shows the overall workflow of the QTPU compiler. At the frontend, we parse the hEinsum into the intermediate representation (QTPU IR). The optimizer rewrites the QTPU IR to balance quantum and classical costs, leveraging both resource modalities. Finally, the backend emits an optimized hybrid kernel.

6.1 Frontend and QTPU IR

The frontend of the compiler converts a given hEinsum program (§ 5.2) to the unified QTPU IR.

qTensor graphs. We convert each qTensor into a graph $G_{\text{QT}} = (V_{\text{QT}}, E_{\text{QT}})$ mirroring the structure of the underlying quantum circuit's TN (Fig. 2), allowing us to reason about which qTensor parts we can offload to classical operations. Each vertex $v \in V_{\text{QT}}$ is a tuple (op_i, q_x), denoting applying operation op_i on qubit q_x. Edges are defined as (i) a *temporal* edge $\big((\text{op}_i, q_x), (\text{op}_k, q_x)\big) \in E_{\text{QT}}$ if op_i immediately precedes op_k on qubit q_x, and (ii) a *spatial* edge $\big((\text{op}_i, q_x), (\text{op}_i, q_y)\big) \in E_{\text{QT}}$ when op_i acts jointly on qubits q_x and q_y.

Operation graph. The operation graph is a directed acyclic

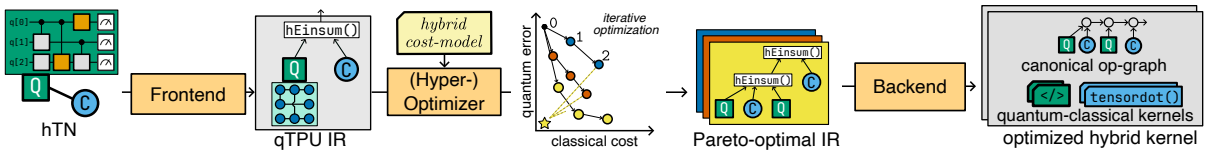


Figure 5: Compiler workflow (§ 6). *The frontend parses hTNs defined with hEinsum to the qTPU IR, the optimizer uses iterative graph partitioning to find Pareto-optimal IR variants, and the backend lowers IR to optimized hybrid quantum-classical kernels.*

graph capturing the order among computations in the hEinsum. Its initial nodes represent qTensors and cTensors, while its middle nodes are hEinsums. After transformations, we obtain a graph where each operation has two operands, allowing us to map each to an efficient classical implementation.

6.2 Hybrid Cost Model

The qTPU compiler optimizes an hEinsum under a dual-objective cost model that balances classical and quantum resources. It captures the classical contraction cost and the expected error of executing quantum circuits on noisy QPUs.

Classical cost. For a contraction hEinsum$(I_1, \ldots, I_n \to O; T_1, \ldots, T_n)$, let $\mathcal{I} = I_1 \cup \cdots \cup I_n$, and let $|i|$ denote the size of index i. The FLOP cost of this contraction is $c_x = \prod_{i \in \mathcal{I} \setminus O} |i|$, i.e., the product of the sizes of all summed-out indices. The total classical cost of the hTN is then $C_{\text{classical}} = \sum_{x \in \text{hTN}} c_x$, the sum of the costs of all nodes in the operation graph.

Quantum cost (error). For each qTensor, we estimate the error of its quantum computation from the structure of its underlying circuit. Let $\mathcal{G}(Q)$ be the set of all quantum operations appearing in qTensor Q, and let $\varepsilon(g)$ denote the error probability of operation g due to noise on the QPU's. We define the per-qTensor error score as $E(Q) = 1 - \prod_{g \in \mathcal{G}(Q)} \big(1 - \varepsilon(g)\big)$. This score approximates the probability that at least one error occurs during the execution of Q. The overall quantum error is the maximum error score over all qTensors.

Optimization goal. The compiler seeks a Pareto-optimal front of points in the joint quantum-classical cost space, balancing classical FLOPs and quantum error to produce the most performant hybrid program for the given hardware.

6.3 Optimization Primitives

We optimize hybrid programs using optimization primitives illustrated in Fig. 6.

Tensorization and decomposition. Tensorization captures a linear combination of related hEinsums as a single hEinsum. As shown in Fig. 6(a), any linear combination of hEinsums differing only by small structural changes can be represented as one hEinsum where coefficients c_i are collected into a classical tensor c, and structural variations are encoded by a qTensor with iswitch operations.

When a qTensor contains separable subcircuits acting on disjoint qubit sets, it decomposes into an equivalent hEinsum of smaller qTensors, since subcircuit measurement outcomes are uncorrelated with the full result obtained by multiplying

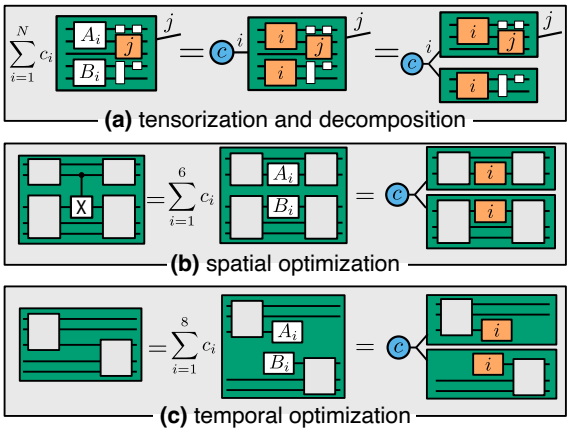


Figure 6: Semantic-preserving qTensor to hEinsum rewrites.

their individual outputs. This replaces one large quantum tensor with several smaller qTensors.

Spatial separation. Gate virtualization partitions circuits by cutting two-qubit gates into a linear combination of single-qubit operations, then recovers the original result by executing all variants and computing their weighted sum [85]. Spatial separation generalizes gate virtualization within our qTPU framework (Fig. 6(b)), replacing the original two-qubit operation with an equivalent hEinsum of one cTensor and one qTensor. When resulting subcircuits operate independently, tensor decomposition factorizes the qTensor into smaller, less noisy components.

Temporal separation. Temporal separation builds on wire cutting, which partitions circuits by cutting qubit wires at specific time steps [80]. The cut decomposes continuous qubit evolution into a linear combination of operations, splitting deep circuits into shallower fragments with reduced accumulated gate errors (Fig. 6 (c)). Applying tensorization and decomposition rewrites large quantum tensors into equivalent hEinsums with multiple shallower, less costly qTensors.

6.4 Optimizer

The optimizer applies the optimization primitives to achieve a Pareto-optimal hybrid cost. It systematically rewrites each qTensor, transforming infeasible quantum workloads into a more balanced hybrid computation.

Optimization workflow. We illustrate the workflow optimizing one qTensor into a hEinsum expression in Fig. 7. We start with the qTensor graph treated as a single partition, and the operation graph containing one qTensor node. At this point, the quantum error is high and the classical cost

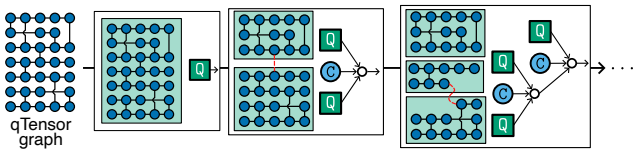


Figure 7: Optimizing a qTensor by iteratively partitioning and recursively building an optimized operation-graph.

is zero, as indicated by the upper-left point in the graph of Fig. 5.

In each iteration, we partition the qTensor graph into $p \geq 2$ balanced subgraphs to minimize cut edges. These boundary edges dictate the optimization primitive: temporal cuts trigger temporal optimization, and spatial cuts trigger spatial optimization (cf. Fig. 6). As shown in Fig. 7, each rewrite replaces the node with an equivalent subgraph of qTensors and cTensors. We greedily iterate by splitting the partition with the highest quantum cost.

The process continues until each rewrite decreases the distance to the Pareto-optimal point. This produces the cost trajectory shown in Fig. 8: each iteration reduces quantum cost (by shrinking qTensors) while increasing classical cost, thereby moving closer to the optimum. Once an iteration increases the distance to this optimum, we terminate.

Hyperparameter optimization. Because the effectiveness of each iteration depends on the chosen partitioning, we run the optimizer with multiple randomized hyperparameter configurations. These hyperparameters include the target partition count per iteration, p, and the relative weight of partition balance versus minimizing inter-partition edges. Each hyperparameter configuration yields a distinct cost trajectory; from these, we select the Pareto-optimal solution, *i.e.,* the one with final cost closest to the theoretical optimum.

Optimization Constraints. The optimizer is subject to three primary optional constraints: (1) the maximum number of qubits per qTensor in the optimized hEinsum, reflecting the available physical resources of the underlying QPUs; (2) an upper bound on the quantum error accumulated within each qTensor; and (3) a cap on the allowable classical post-processing cost, measured in FLOPs. When such constraints are specified, we first restrict the solution space to Pareto-optimal candidates that satisfy all constraints, and then select the remaining Pareto-optimal solution.

6.5 Backend

Finally, the qTPU compiler lowers the optimized qTPU IR into concrete hybrid kernels.

Contraction scheduling. The operation graph is canonicalized so that every node has exactly two operands, ensuring that each contraction corresponds to a single accelerator operation that can efficiently be implemented on a classical device. We do so while optimizing the contraction order.

Lowering qTensors. Each qTensor across all slices is lowered into a single parametric quantum kernel. For every symbolic index of the qTensor, the compiler introduces a parameter, and each iswitch annotation becomes a corresponding `if-else` block that selects the appropriate gates.

7 qTPU Runtime

The qTPU runtime enables adaptable, scalable, and highly parallel execution of optimized hEinsum programs across a heterogeneous set of quantum and classical accelerators.

Workflow. Fig. 8 illustrates the workflow of the qTPU runtime. The runtime receives the compiler-optimized hybrid kernel, and execution proceeds in two conceptual phases.

In the *adaptation phase* (Fig. 8 (a)), the runtime adapts and lowers the hEinsum expression to the currently available resources. The index slicer and device placer partition the expression into multiple independent slices to expose parallel work and assign each slice's quantum and classical operations to specific QPUs or classical accelerators. After placement, the code generator compiles the mapped quantum and classical operations into device-specific kernel code.

In the *execution phase* (Fig. 8 (b)), the runtime executes the sliced hEinsum across the distributed devices. The runtime first allocates all qTensors and cTensors and loads the generated kernels onto their assigned hardware. Execution proceeds in a map–reduce style: the qTensor engine evaluates all qTensors in parallel on QPUs, the cTensor engine contracts the resulting slices on classical accelerators, and the partial contractions are then combined into the result.

7.1 Index Slicing and Device Placement

During the adaptation phase (Fig. 8 (a)), the slicer and device placer prepare the optimized hEinsum for distributed execution across QPUs and classical accelerators, with the goal being to expose as much parallel quantum work as possible, as QPUs are typically the throughput bottleneck.

Let i be an index with domain $\{0, \ldots, |i|-1\}$. Slicing along i simply means creating one hEinsum for each value of i, *i.e.,* a set of sub-expressions $\mathsf{hEinsum}(\text{expr}; T_1, \ldots, T_n)\big|_{i=v}$. Each of these subproblems can be evaluated independently, and the original result is recovered by summing their outputs:

$$\mathsf{hEinsum}(\text{expr}; T_1, \ldots, T_n) = \sum_{v=0}^{|i|-1} \mathsf{hEinsum}(\text{expr}; T_1, \ldots, T_n)\big|_{i=v}.$$

In the example of Fig. 8, we slice an index with $|i| = 2$ at qTensor A, giving us two distinct slices.

To generate sufficient QPU parallelism, the slicer iteratively selects indices adjacent to each qTensor and slices along those that maximize the number of independent circuit instances, continuing until at least as many qTensor instances are produced as there are available QPUs.

After slicing, the device placer assigns each qTensor instance to a QPU with available capacity, using a load-balanced assignment when the number of instances exceeds the devices. The remaining classical slice contractions are

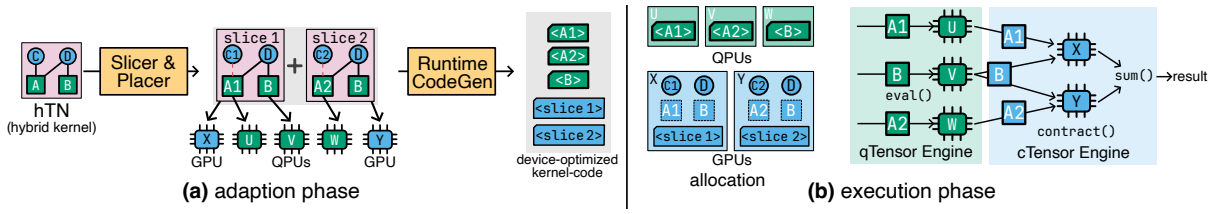


Figure 8: qTPU runtime (§ 7).*(a) Adaptation phase: the slicer and placer partition the hTN into independent slices assigned to QPUs and GPUs, then the code generator compiles device-specific kernels. (b) Execution phase: the qTensor engine evaluates quantum circuits in parallel on QPUs, the cTensor engine contracts results on classical devices to produce the final result.*

mapped to classical accelerators in a round-robin manner.

7.2 Runtime Code Generation

Following device placement, the runtime code generator compiles the hybrid kernel into device-specific executables: for every qTensor assigned to a QPU, it emits a quantum kernel using the native toolchain, while for classical slices, it compiles the shared operation graph into a single optimized kernel for the target accelerator using a tensor backend [15].

7.3 Tensor Engines

The tensor engines execute the sliced hEinsum across QPUs and classical accelerators.

The **qTensor engine** evaluates all qTensors on available QPUs. For each unique qTensor, it allocates quantum resources and the output buffer for its result cTensor entry, then loads the compiled quantum kernel onto the selected QPU. The compiled kernel is executed for each qTensor instance, returning results one by one into a result-cTensor.

The **cTensor engine** performs classical contractions for each slice. For every slice, it allocates slice-local cTensor buffers on the assigned classical accelerator and loads the compiled contraction kernel from the code generator. The engine waits asynchronously for cTensor inputs from the qTensor engine; once all required inputs arrive, it launches the slice's contraction kernel. Each slice produces a partial result, which the cTensor engine reduces across the sliced indices to yield the final output.

7.4 Fault Tolerance

The runtime handles failures gracefully, since QPUs and classical devices may become unavailable during execution. The sliced hEinsum structure naturally supports fault tolerance, as each slice represents an independent unit of work.

QPU calibrations. Unlike classical accelerators, QPUs require regular calibration to maintain low noise, typically going offline for about an hour daily [1]. The qTPU runtime models calibration as scheduled downtime: when a QPU enters calibration, the qTensor engine completes any in-flight circuit (usually seconds) and stops dispatching new qTensor instances to that device. Pending instances are reassigned to remaining QPUs using the same load-balanced policy.

Classical failures. If a classical accelerator fails during cTensor contraction, the cTensor engine re-executes only affected slice contractions. Since each slice depends solely on input cTensors (already in host memory), recovery requires no cross-slice coordination. Uncompleted contractions are reassigned to surviving accelerators.

8 Implementation and Evaluation

8.1 System Implementation

We implement an end-to-end qTPU prototype in Python.

Programming Model. We provide Python constructs for hEinsum and qTensors. To declare a qTensor, users annotate a Qiskit [68] quantum circuit with our custom iswitch operations. A hEinsum operation is then constructed with a einsum expression and corresponding qTensor objects and classical PyTorch [60] tensors.

Compiler. The compiler uses KaHyPar 1.3 [75] for graph partitioning at each recursive optimization level and explores the Pareto frontier of classical cost versus quantum error using randomized multi-start optimization. The backend returns a cotengra [34] operation graph and generates qTensors as CUDA-Q [18] kernels.

Runtime. The runtime leverages cotengra as an interface for contraction optimization and slicing, with PyTorch for tensor operations. Our code generator JIT-compiles qTensors into optimized QPU-native code using CUDA-Q, and classical operations into GPU kernels using the PyTorch toolchain.

8.2 Hybrid Application Implementation

To measure qTPU's performance on real-world hybrid applications, we implement three real-world applications in the qTPU programming model, enabling case studies on each (§ 8.6–§ 8.8): (i) for **hybrid ML**, we implement the example from Lst. 2, (ii) for **scalable hybrid computation**, we implement circuit knitting by decomposing a quantum circuit using the state-of-the-art circuit knitting library – QAC [11] and parsing the cut circuit into an equivalent hEinsum, and (iii) for **error mitigation**, we adopt probabilistic error cancellation (PEC) and Pauli twirling from the state-of-the-art error-mitigation library Mitiq [47], and convert them into an hEinsum.

8.3 Experimental Methodology

Experimental setup. We use a server with two AMD EPYC 9654 96-Core processors (384 cores with HT), 1.5 TB RAM,

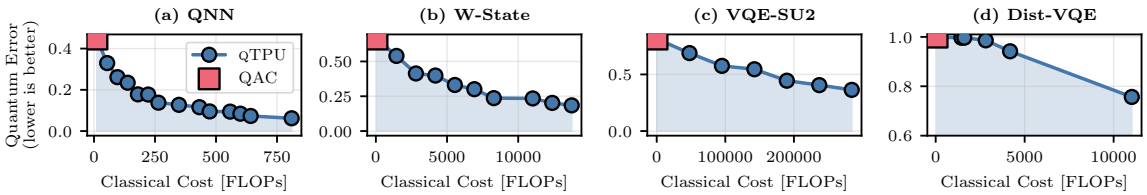


Figure 9: Compiler tradeoff (§ 8.4). *QAC produces a single solution, while QTPU exposes a flexible Pareto frontier.*

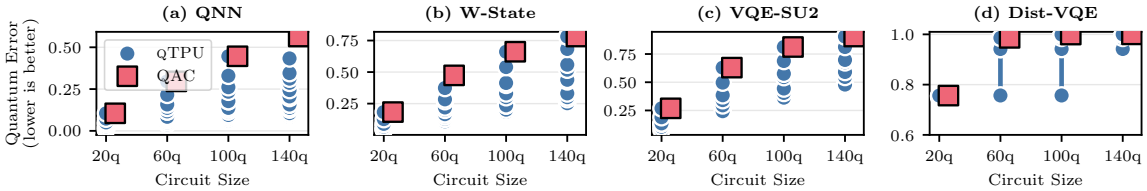


Figure 10: Compiler optimality analysis (§ 8.4). *While QAC yields a single solution with error scaling linearly with circuit size, QTPU maintains consistent error reduction across scale.*

	20q	40q	60q	80q	100q	120q	140q
QTPU [s]	0.76	1.05	1.25	1.77	2.01	2.34	2.51
QAC [s]	1.2	10.9	35.0	69.3	91.5	115.0	134.1
Speedup	**1.5×**	**10.3×**	**28.0×**	**39.2×**	**45.5×**	**49.1×**	**53.4×**

Table 1: Compile time scaling for VQE-SU2 circuits. QTPU's compile time is near-constant; QAC scales superlinearly.

and an A40 GPU with 48GB HBM3 memory.

Baselines. We evaluate QTPU against Qiskit-Addon-Cutting (QAC) [11], the current state-of-the-art framework for hybrid quantum-classical execution through circuit decomposition. We compare our runtime against cuTensorNet [56], NVIDIA's GPU-accelerated library for circuit simulation, to establish a classical-only baseline. For the error-mitigation case-study, we compare against MITIQ [47], the leading framework for quantum error mitigation. For our hybrid ML case-study, we compare against the BATCH method (Lst. 1), which is commonly used in quantum frameworks [9, 68]

Benchmarks. We evaluate four quantum circuit families from MQT Bench [70], representative of hybrid quantum-classical applications: **QNN**, a quantum neural network for quantum machine learning [66]; **W-State**, preparing the n-qubit linearly entangled W-state [25]; **VQE-SU2**, a Variational Quantum Eigensolver ansatz with efficient $SU(2)$ rotations for quantum chemistry [63, 67]; and **Dist-VQE**, a distributed VQE variant with clustered connectivity modeling natural circuit partitioning across multiple QPUs.

Metrics. We evaluate qTPU using the following metrics: (1) **Quantum error**: The estimated probability of at least one error occurring during circuit execution (§ 6.2). We assume a uniform error model with 10^{-3} for single-qubit gates and 10^{-2} for two-qubit gates [1]. (2) **Classical cost**: The number of floating-point operations (FLOPs) required for classical computation (§ 6.2). (3) **Runtime**: Wall-clock execution time in seconds, measured either for classical or quantum or end-to-end hybrid execution. (4) **Code size**: The number of lines of generated CUDA-Q quantum-kernel code. (5) **QPU time**: Estimated quantum hardware execution time, computed by transpiling circuits to IBM Marrakesh using Qiskit's ASAP scheduling with optimization level 3 using 1000 shots [1, 68].

8.4 QTPU Compiler

RQ1 (Compiler trade-off): *Can the compiler trade quantum error for classical computation?* We partition each 100-qubit benchmark into 50-qubit subcircuits using both QAC and QTPU, with QTPU configured to run 50 hyperparameter trials, retaining only Pareto-optimal solutions.

Fig. 9 shows that while QAC produces a single fixed solution, QTPU offers multiple configurations trading classical cost for quantum error. Mid-range solutions yield 1.5–3.4× error reduction, while maximum classical investment achieves 2.2–7.2× improvement across all benchmarks. At equal classical cost, both frameworks provide the same solution-quality

RQ2 (Compiler scalability): *Does the compiler scale to large qTensors?* We vary the circuit size from 20 to 140 qubits, cutting each circuit into subcircuits of 50% the original size.

Fig. 10 shows QTPU maintains stable error reduction across scales, ranging from 2–26× depending on the benchmark. While QAC error scales linearly with circuit growth, QTPU's Pareto frontiers consistently trade FLOPs for reduced quantum error without diminishing returns.

Table 1 shows the compile time on VQE-SU2: QAC scales superlinearly (1.2s to 134s) due to exhaustive search, while QTPU maintains ~1–3s (up to 53× speedup for 140 qubits).

8.5 QTPU Runtime

RQ3 (Runtime analysis): *What is QTPU's runtime performance and where is it spent?* We measure total runtime, including compilation, classical postprocessing, and QPU execution, on a 15-qubit QPU, requiring the compiler to optimize qTensors to a circuit size of 15 qubits or fewer.

Fig. 11(a) shows runtime scaling with circuit size across benchmarks. For 100-qubit circuits, quantum execution dominates at 75-100% of total runtime, while classical contraction accounts for <0.01% and compilation takes 2-3s.

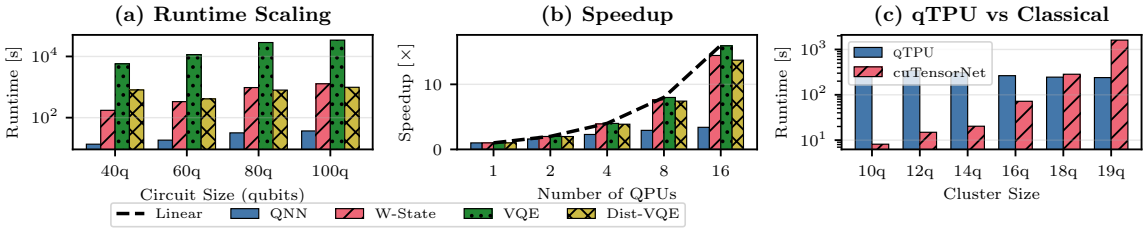


Figure 11: qTPU runtime analysis (§ 8.5). *(a) Total runtime scales with circuit size across benchmarks. (b) Near-linear speedup with multiple QPUs (14.4× at 16 QPUs for W-State) due to embarrassingly parallel qTensor execution. (c) qTPU vs. cuTensorNet on 100q Dist-VQE: qTPU outperforms classical beyond 18-qubit clusters, achieving 6.7× speedup at 19-qubit clusters.*

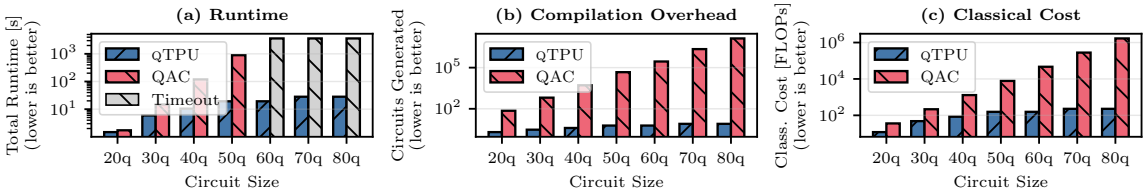


Figure 12: Scalable hybrid computing for QNN on a 10-qubit QPU (§ 8.7). We partition QNN circuits (20–80 qubits) into 10-qubit subcircuits using qTPU and QAC. *(a) qTPU scales to 80 qubits in <30s while QAC times out beyond 50 qubits. (b) qTPU generates 10–42× fewer subcircuits. (c) qTPU's classical postprocessing requires* 10^2–10^3 *FLOPs vs. QAC's* 10^6 *FLOPs.*

RQ4 (Scalability): *Can the runtime leverage multiple QPUs?* Fig. 11(b) shows near-linear speedup as QPU count increases from 1 to 16, achieving 14.4× speedup with 16 QPUs (90% efficiency). The slicer produces embarrassingly parallel qTensor instances, thereby minimizing distribution overhead.

RQ5 (Comparison to classical simulation): *How does qTPU perform against pure classical simulation?* We evaluate 100-qubit Dist-VQE circuits with clustered connectivity. qTPU executes on 4 QPUs with 1,000 shots per subcircuit; cuTensorNet performs full simulation.

Fig. 11 (c) shows that as cluster size increases, qTPU runtime decreases (fewer qTensors and cTensors) while cuTensorNet runtime grows exponentially (larger clusters yield harder contractions). qTPU outperforms classical beyond 18-qubit clusters, achieving 6.7× speedup at 19-qubits.

8.6 Case-Study: Hybrid Machine Learning

We compare manual batching (Lst. 1) against qTPU's unified model (Lst. 2) for evaluating hybrid quantum-classical ML. We use QNN quantum kernels of 20–100 qubits with $X \in \mathbb{R}^{b \times h}$, $V \in \mathbb{R}^{h \times f}$, batch sizes $b \in \{50, 100, 200\}$, hidden dimension $h = 20$, and feature dimension $f = 8$.

RQ6 (Compilation Scalability): *How does compilation time scale with problem size?* Fig. 13 (a,b) shows compilation scaling with circuit and batch size. qTPU achieves 3.7× average speedup (up to 6×) over batch execution. Critically, qTPU scales sublinearly with batch size, whereas batch execution scales linearly by compiling each circuit independently. At 100 qubits, batch execution grows from 3.1s to 63.5s as batch size increases from 10 to 200, while qTPU grows only from 4.8s to 16.9s, a speedup of up to 4.5×.

RQ7 (Code Generation Overhead): *What is the impact on generated code size?* Fig. 13 (c,d) shows qTPU's code generation efficiency. By compiling the quantum kernel once with parametric control flow rather than instantiating separate circuits, qTPU achieves an average reduction of 33× (up to 48×). At 100 qubits with batch size 200, qTPU generates 132k lines vs. batch execution's 5.1M lines—a 38.7× reduction. This directly translates into faster compilation and a smaller memory footprint, with benefits that scale with batch size.

8.7 Case Study: Scalable Hybrid Computing

In this case study, we implement scalable hybrid computing using circuit knitting on resource-constrained hardware. We run qnn benchmarks on 20-80 qubits, targeting a 10-qubit QPU and enforcing aggressive circuit partitioning.

RQ8 (End-to-end runtime): *What impact does qTPU have on the end-to-end runtime with increasing problem size?* Fig. 12 (a) shows total runtime: QAC scales exponentially and times out beyond 40 qubits (20-minute limit), while qTPU completes 80-qubit circuits in under 30 seconds. This dramatic difference stems from qTPU's hTN representation, avoiding redundancies by compactly encoding the entire computation.

RQ9 (Compilation overhead): *How many circuits does qTPU generate with increasing problem size?* Fig. 12 (b) reveals the root cause of QAC 's exponential scaling: circuit generation grows combinatorially (12 to 6,480 circuits from 20q to 50q), whereas qTPU produces 10–42× fewer subcircuit instances. The hTN abstraction eliminates this overhead by representing subcircuits declaratively as qTensors rather than enumerating explicit circuit variants.

RQ10 (Postprocessing overhead): *What is the classical cost in FLOPs that qTPU incurs with increasing problem size?* Fig. 12 (c) shows the classical postprocessing cost: qTPU maintains modest overhead ($< 10^3$ FLOPs) across all sizes, while QAC grows to 10^6 FLOPs exponentially. The hTN formulation performs only the minimum necessary tensor contractions by structure-aware optimization, whereas QAC's

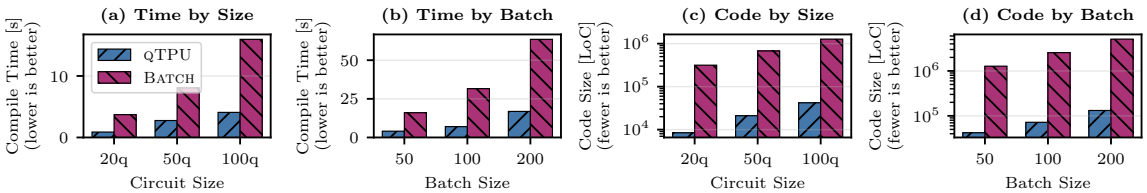


Figure 13: Case study: hybrid machine learning with BATCH execution (Lst. 1) vs. QTPU (Lst. 2) (§ 8.6). *(a) Compilation time by circuit size, (b) compilation time by batch size, (c) generated code size by circuit size, (d) generated code size by batch size.*

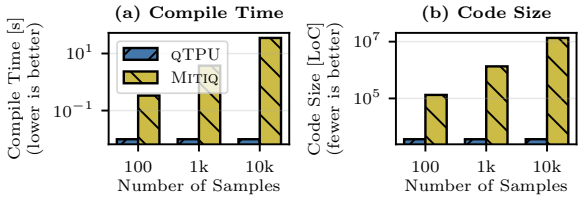


Figure 14: Quantum error mitigation (QEM) vs MITIQ the 100-qubit QNN benchmark (§ 8.8). ***(a)*** *QTPU is 3,500× faster in compilation.* ***(b)*** *QTPU generates 3,700× fewer lines of code.*

explicit circuit enumeration incurs exponential classical reconstruction overhead proportional to the circuit size.

8.8 Case Study: Quantum Error Mitigation (QEM)

RQ11 (QEM scalability): *What impact does QTPU have on the compile time and code size requirements of a hybrid QEM workflow?* We implement three major error mitigation strategies (§ 8.2) using both QTPU and Mitiq on quantum circuits with 200 single-qubit gates. For Mitiq, we vary the sample count from 100 to 10,000 circuits; QTPU represents the full task symbolically as a hEinsum, independent of sample count.

Fig. 14 compares compile-time overhead and code size. While Mitiq explicitly generates each circuit variant, QTPU represents the entire ensemble symbolically, encoding up to $4^{200} \approx 10^{120}$ configurations (for combined PEC+Twirling applied to all gates) as a single qTensor. QTPU maintains constant ~10ms overhead regardless of sample count, while Mitiq scales linearly from ~335ms at 100 samples to 35s at 10,000 samples (~3,550×). Similarly, QTPU generates only ~3.7k lines of code, whereas Mitiq generates 13.5M (~3,700×).

9 Related Work

Hybrid quantum-classical applications. A wide range of quantum applications inherently rely on hybrid programming and execution models. Variational algorithms [27, 63], quantum machine learning [10, 49], quantum error mitigation [13, 20–22, 26, 62, 81, 82, 84, 92], and hybrid runtime systems [32,33,80,83,85] all interleave quantum circuit execution with classical optimization and/or postprocessing.

However, current approaches remain constrained by fragmented programming models, a lack of holistic optimizations, and a lack of resource-adaptable execution. In contrast, QTPU provides a unifying abstraction based on hTNs that encode entire hybrid computations. This enables global optimization, hardware-aware compilation, and flexible execution across QPUs and classical accelerators.

Tensor networks (TNs) in quantum simulation. TN-based quantum circuit simulation represents quantum gates and states as tensors, where simulation is performed through systematic tensor contraction operations [7, 54, 59, 73, 76, 89–91]. In contrast, QTPU employs tensor networks to enable hybrid execution across both quantum and classical accelerators, where quantum tensors are physically executed on QPUs rather than simulated, allowing portions of the computation to leverage real quantum hardware while classical tensors exploit GPU/TPU acceleration.

Tensor optimization in classical systems. Deep learning compilers optimize tensor operators through scheduling transformations, operator fusion, and hardware mapping to maximize throughput on classical accelerators [15, 23, 41, 53, 78, 87, 94–96]. Similarly, general tensor compilers focus on efficient contraction and algebraic manipulation of dense and sparse tensors, addressing memory layouts, parallelization, and sparsity exploitation for scientific computing and machine learning workloads [6, 24, 31, 40, 44–46, 88]. QTPU addresses fundamentally different challenges: rather than optimizing dataflow graphs of deterministic, fully materialized tensor operations, QTPU optimizes hybrid tensor *network* contractions that integrate quantum and classical processing, requiring the compiler to determine an optimal trade-off between the two compute paradigms.

10 Conclusion

We introduce the hybrid tensor network (hTN) abstraction, a unified framework that seamlessly integrates quantum and classical computation. hTNs address current hybrid computing limitations—programming fragmentation, static partitioning, and poor scalability—through holistic representation and optimization. Our system, QTPU, realizes hTNs through a declarative programming model, a compiler that balances quantum error with classical cost, and an adaptable runtime for heterogeneous hybrid quantum-classical resources.

Evaluation across hybrid ML, circuit knitting, and error mitigation demonstrates that QTPU achieves up to 53× faster compilation, 7.2× lower quantum error rate, 3,700× code reduction, and 10,000× lower classical overhead than state-of-the-art baselines. Through its concise, flexible, and scalable abstraction for quantum-classical acceleration, QTPU enables automated and efficient heterogeneous resource utilization for next-generation quantum-enhanced applications.

Artifact. QTPU will be publicly available.

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